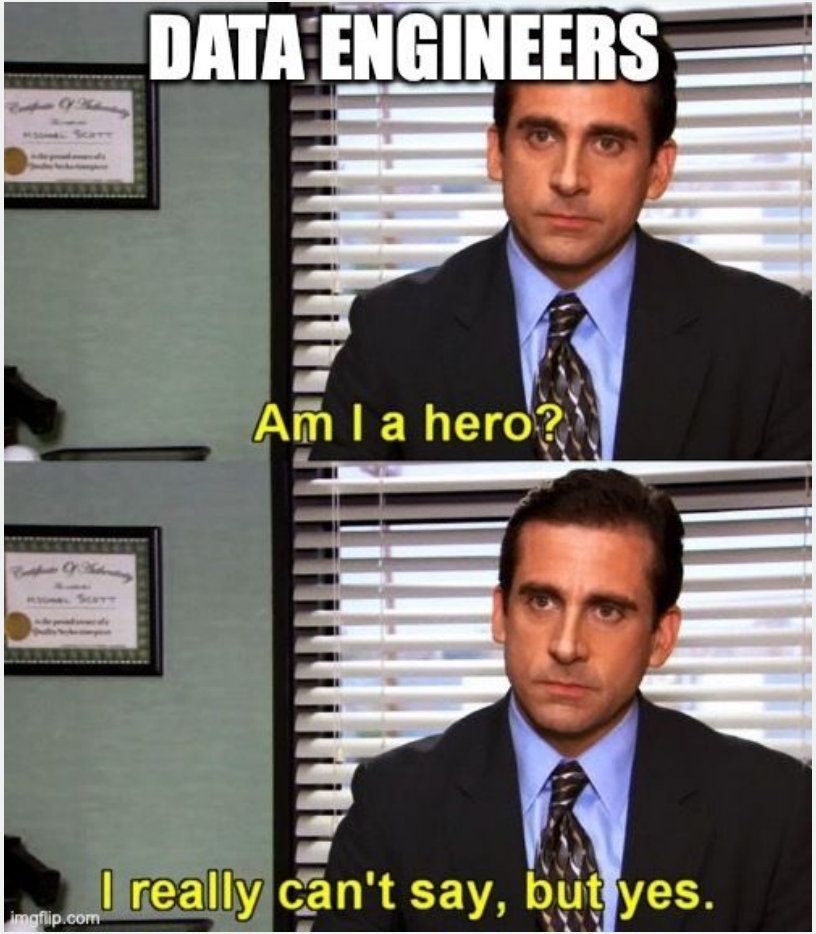
LAB #11

Adv. CiC
Spring 2026

Agenda

1. Housekeeping
2. Performance, cont.
3. Data Pipelines
4. Workflows
5. Lab 11 Exercise



Housekeeping

Due this past week

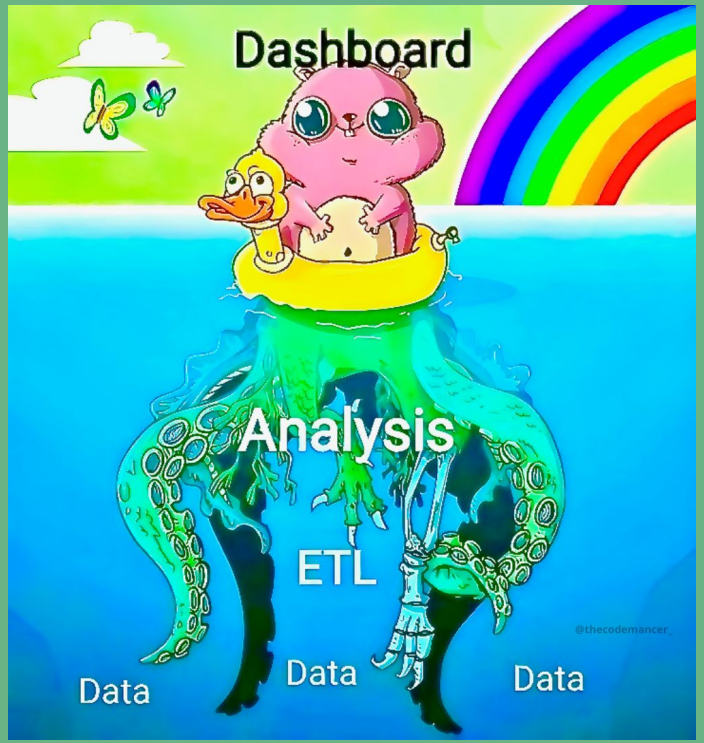
- Project check-in
- Lab 10

Upcoming

- Readings - **4/9 @ 3pm**
- Lab 11 - **4/10 @ 1pm**

Announcements

1. Grading catch-up (sorry!)
2. Project feedback



Questions

(on Lectures, Labs, or Readings)?

Performance

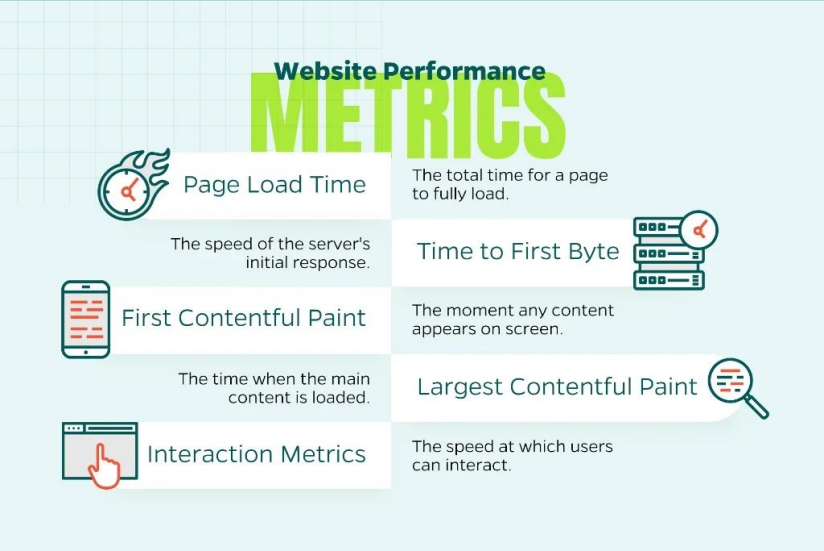
1. Slow apps **frustrate users** and drive abandonment.

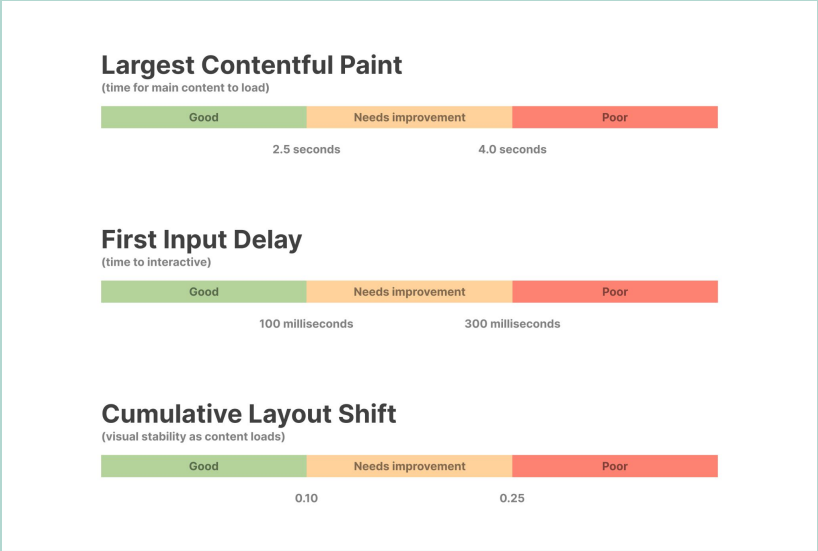
2. Fast interfaces feel polished and **build trust**.

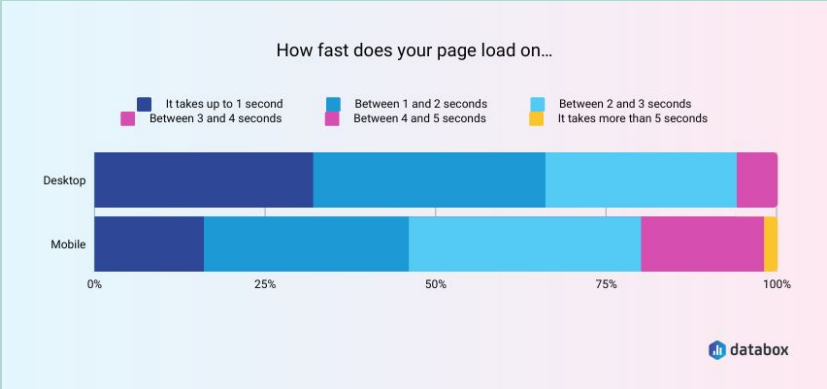
3. Performance directly **impacts conversion** and retention.

4. Scalable apps **handle growth** without degrading experience.

5. Poor performance **excludes users** on low-end devices or slow networks.

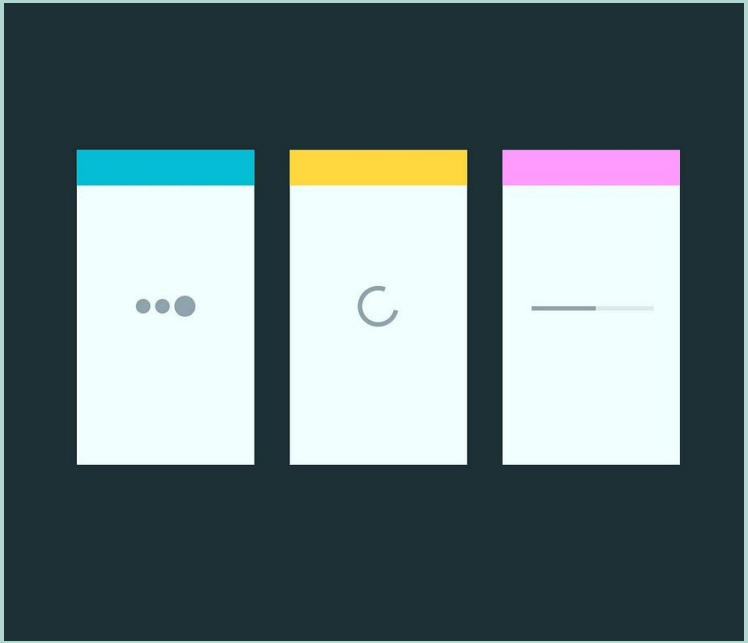

Website Performance
METRICS
Page Load Time
The total time for a page to fully load.
The speed of the server's initial response.
Time to First Byte
First Contentful Paint
The moment any content appears on screen.
The time when the main content is loaded.
Largest Contentful Paint
Interaction Metrics
The speed at which users can interact.

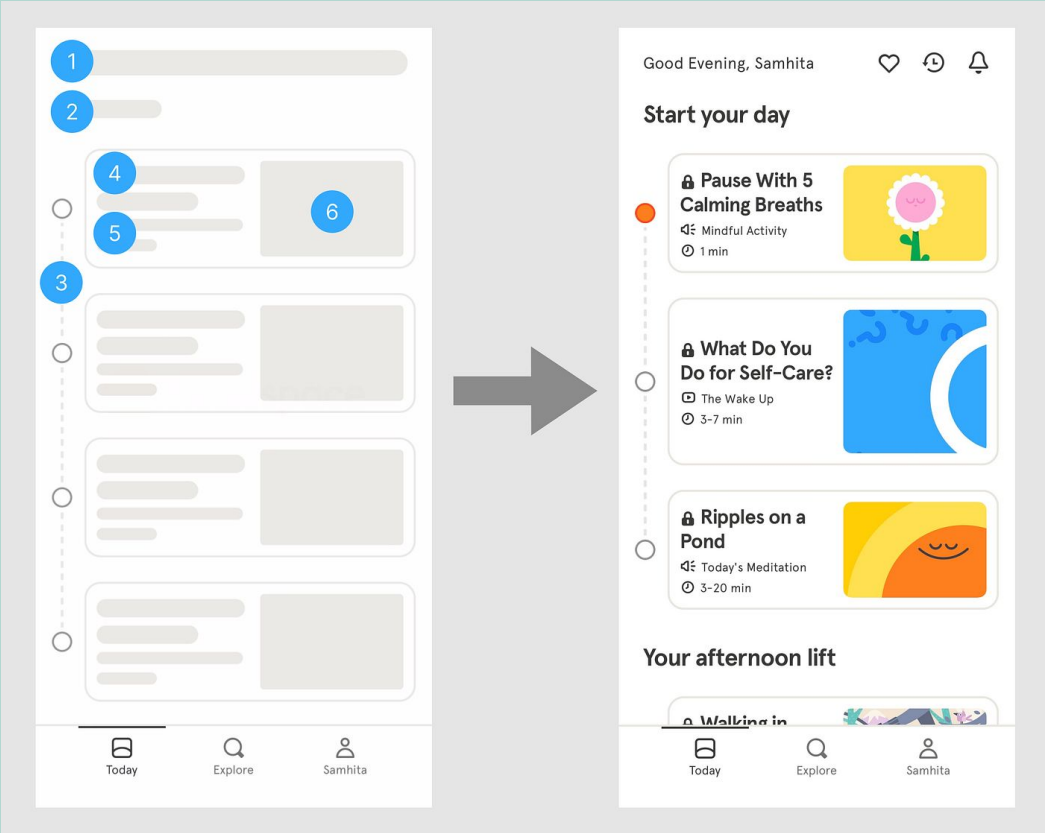

Largest Contentful Paint
(time for main content to load)
Good
Needs improvement
Poor
2.5 seconds
4.0 seconds
First Input Delay
(time to interactive)
Good
Needs improvement
Poor
100 milliseconds
300 milliseconds
Cumulative Layout Shift
(visual stability as content loads)
Good
Needs improvement
Poor
0.10
0.25


How fast does your page load on...
It takes up to 1 second
Between 1 and 2 seconds
Between 2 and 3 seconds
Between 3 and 4 seconds
Between 4 and 5 seconds
It takes more than 5 seconds
Desktop
Mobile
0%
25%
50%
75%
100%
databox

Improvements

1. **Reduce data transfer size** — optimize your SQL queries and keep payloads lean.
2. **Clean up your code** — remove unnecessary or slow steps so it runs faster and is easier to maintain.
3. **Be strategic about how things load** — parallel or lazy loading prevent bottlenecks and keep the experience smooth.
4. **Cache what you can** — storing frequently accessed data cuts down on redundant fetches.
5. Use **UX mitigations** as needed — visual elements such as progress bars, spinners, and skeleton views manage perceived performance. Even if something takes time, users should never feel stuck.




1
2
3
4
5
6
Today
Explore
Samhita
Good Evening, Samhita
Start your day
Pause With 5 Calming Breaths
Mindful Activity
1 min
What Do You Do for Self-Care?
The Wake Up
3-7 min
Ripples on a Pond
Today's Meditation
3-20 min
Your afternoon lift
Today
Explore
Samhita

Drive

Search Drive

Clio

New

My Drive

Team Drives

Shared with me

Recent

Starred

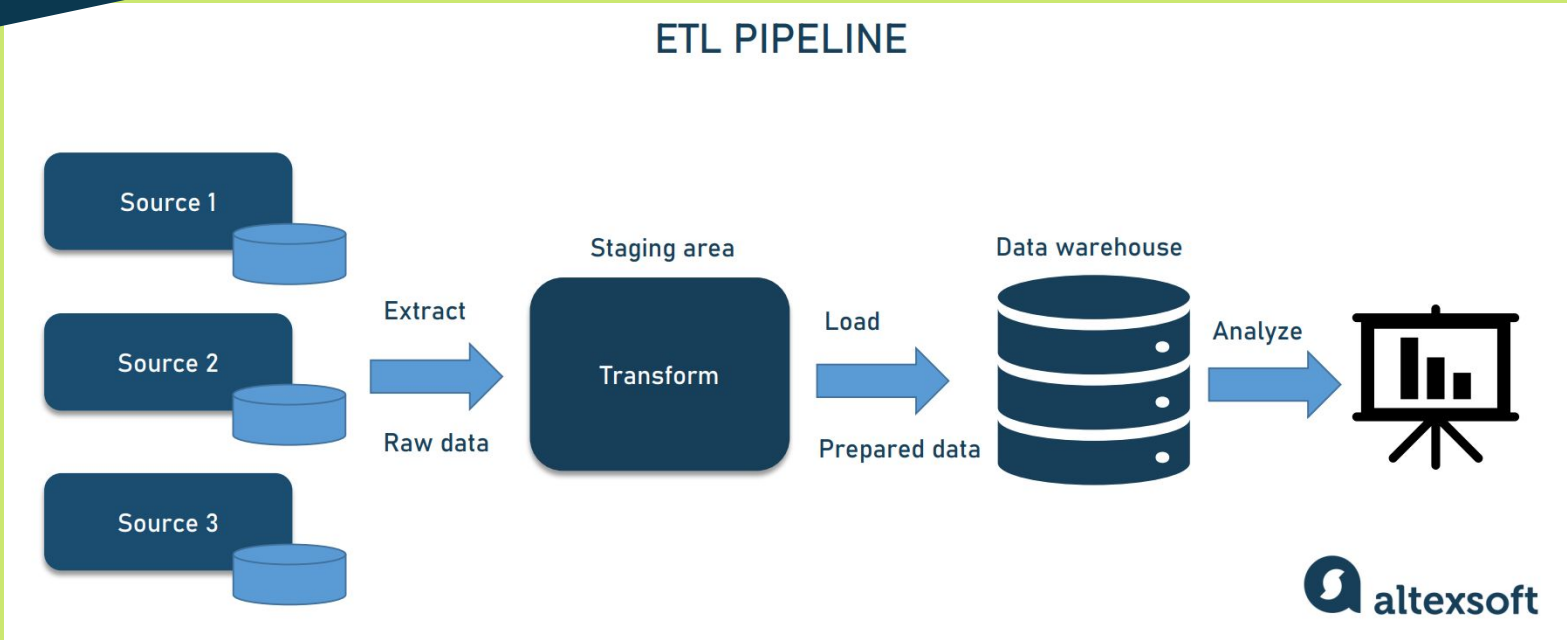
Trash

Quick Access

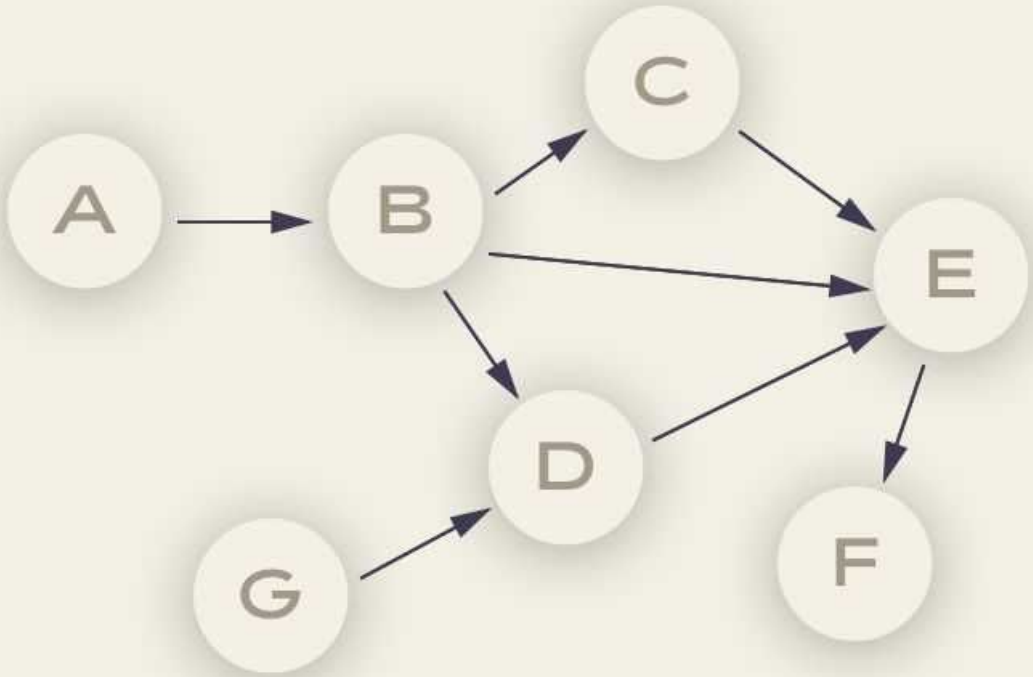
Name	Owner	Last modified	Size
Accounting	me	May 25, 2018 me	–
Delivery	me	May 25, 2018 me	–
Design System	me	Jun 29, 2018 me	–
DesignCraft	me	Jun 29, 2018 me	–
General	me	May 25, 2018 me	–
Inception	me	Jul 26, 2018 me	–

Data Pipelines

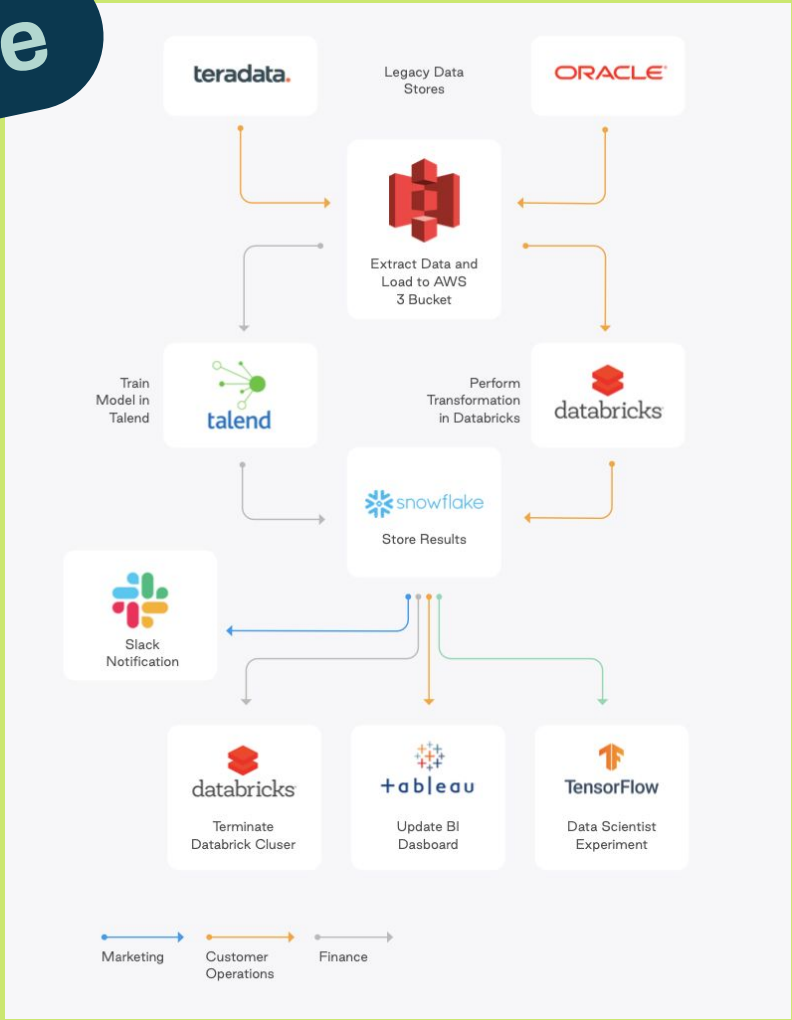
ETL


ETL PIPELINE
Source 1
Source 2
Source 3
Extract
Raw data
Staging area
Transform
Load
Prepared data
Data warehouse
Analyze
altexsoft

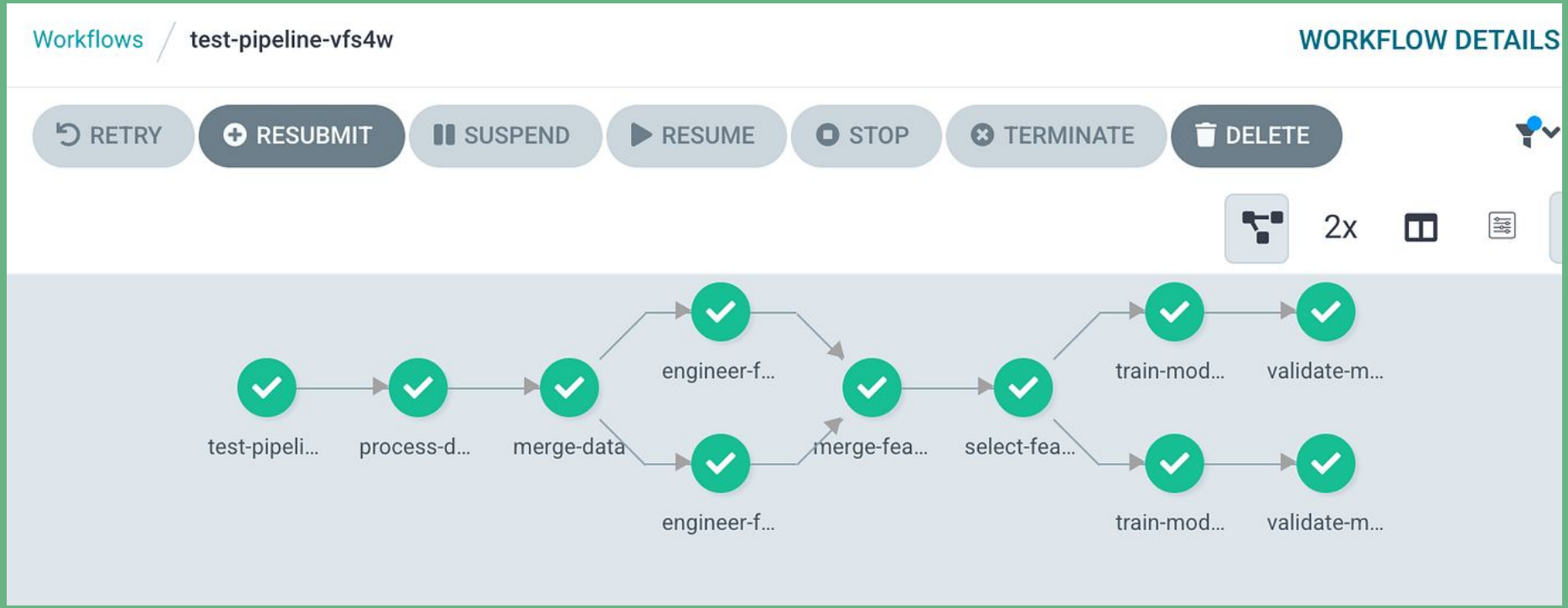
DAG



DAG Example


teradata.
Legacy Data Stores
ORACLE
Extract Data and Load to AWS 3 Bucket
Train Model in Talend
talend
Perform Transformation in Databricks
databricks
snowflake
Store Results
Slack Notification
databricks
Terminate Databrick Cluser
tableau
Update BI Dasboard
TensorFlow
Data Scientist Experiment
Marketing
Customer Operations
Finance

Workflows



A workflow is a **defined, automated process (sequence of steps)** that will run one or more jobs.

github.com/advanced-computing/course-materials/actions

Finish update

advanced-computing / course-materials

Type / to search

Code | Issues | Pull requests | Actions | Security and quality | Insights | Settings

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.

Actions

New workflow

All workflows

Automatic Dependency Submission
Check for broken links
Copilot coding agent
Dependabot Updates
Run tests
Update Microsoft Stock Price Data

Management

Caches
Attestations
Runners
Usage metrics
Performance metrics

All workflows

Showing runs from all workflows

Filter workflow runs

Help us improve GitHub Actions
Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback

757 workflow runs

Workflow | Event | Status | Branch | Actor

Automatic Dependency Submission (Python) Automatic Dependency Submission #205: by github-advanced-security bot	main	39 minutes ago 1m 18s	...
Add files via upload Run tests #435: Commit ad27bf1 pushed by spalle1997	main	39 minutes ago 1m 0s	...
Automatic Dependency Submission (Python) Automatic Dependency Submission #204: by github-advanced-security bot	main	Apr 2, 11:29 PM EDT 1m 14s	...
add caching demo Run tests #434: Commit 99a136b pushed by afeld	main	Apr 2, 11:29 PM EDT 58s	...
clarify performance expectations Run tests #433: Commit 686e57d pushed by afeld	main	Apr 2, 4:08 PM EDT 1m 1s	...
Automatic Dependency Submission (Python) Automatic Dependency Submission #203: by github-advanced-security bot	main	Apr 2, 4:08 PM EDT 1m 12s	...

UFC
00:02:42
UFC
UFC

EXERCISE

Lfg